

FR-016 M3 — Minimal Owner Workflow for Learning Proof

FR016 — `sources/optional/m3.md`

Masterclass source package snapshot

Functional Requirement	FR-016
Edition	Engineering Learning Academy — Source Package snapshot
Status	Complete (M3) — **historical milestone record
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python `scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

Contents

FR-016 M3 — Minimal Owner Workflow for Learning Proof

Date: 2026-08-06

Status: Complete (M3) — **historical milestone record**

Succeeded by: [M4: acceptance](#) (FR-016 Complete / Frozen)

Architecture: [ADR-008](#)

Preceding: [M2 runtime / go-no-go](#) (**GO AS LEARNING PROOF ONLY**)

Did not begin in M3: M4 freeze, FR-017, product expansion, study-aid generation (M4 later completed the freeze; this report remains the M3 historical record)

1. Executive summary

M3 makes the M2 multi-agent learning proof **minimally owner-operable** under:

```
cic agent orchestrate ...
```

Owner can run / resume / show / history / list bounded orchestration journeys and see specialist selection, authority, typed handoffs, policy results, child AgentRun / OBS brief refs, stop reasons, and next actions.

Binding M2 verdict remains explicit in every report: FR-016 is a learning proof and future substrate — **not** the preferred daily replacement for `cic agent run`.

No new specialists, no frameworks, no LLM supervisor, no discovery/submit/pipeline authority.

2. CLI design

Extended existing `cic agent` (consistent with FR-015) rather than a top-level `cic orchestrate` namespace.

Command	Purpose
<code>cic agent orchestrate run <opp> --goal <g> --approve</code>	Start DOS
<code>cic agent orchestrate resume <orr_...> --approve</code>	Resume after SoT re-inspect
<code>cic agent orchestrate show <orr_...></code>	Owner report
<code>cic agent orchestrate history <orr_...></code>	Append-only audit
<code>cic agent orchestrate list</code>	List runs
<code>cic agent orchestrate check-delegation ...</code>	Teaching: policy admit/deny without execute

Shared path overrides mirror `cic agent ( --dir , --packages-dir , --agent-runs-dir , --orchestration-runs-dir , ...)`.

`--approve` is mandatory for `run / resume`.

3. Supported owner goals

--goal	Maps to	Behaviour
<code>brief</code>	<code>brief_opportunity_readiness</code>	DOS → OBS
<code>prepare</code>	<code>coordinate</code> (no <code>synthesize</code>)	DOS → BOPA
<code>prepare_then_brief</code>	<code>coordinate</code> + <code>synthesize_after_prepare</code>	DOS → BOPA → OBS

No additional goals or specialists.

4. Implementation summary

Piece	Location
Goals	<code>career_intelligence.multi_agent.goals</code>
Factory	<code>build_orchestration_supervisor</code>
Presentation	<code>presentation.py</code> (M3 owner report)
CLI	<code>cic agent orchestrate</code> in <code>cli/main.py</code>
Store	<code>data/orchestration_runs/</code>

BOPA allow-list / ToolPolicy unchanged.

5–7. Presentation (selection, authority, handoff/audit)

`show` prints:

1. Learning-proof banner + preference for `cic agent run`
2. Owner goal + opportunity + status / stop / steps
3. Parent / child IDs (orchestration, BOPA AgentRun, OBS brief)
4. Observed derived state
5. Specialist visits
6. Per-handoff: selection reason, delegation policy, lifecycle, **authority boundary** (BOPA vs OBS allow-list summary), child result refs
7. Owner action mapped to legal next command
8. Safety footer (no submit / pipeline / truth waive / discovery)

`history` lists append-only orchestration events.

8. Resume behaviour

Unchanged from M2: re-inspect SoT; skip unchanged OBS brief; BOPA child resume / idempotent prep; `--approve` required; global limits preserved.

9. Manual validation

`python scripts/run_fr016_m3_manual.py` covers A–I (offline DOS + CLI approve gate + illegal delegation deny).

ID	Journey	Expected
A	brief	OBS briefing
B	prepare	BOPA → owner stop
C	prepare_then_brief	BOPA → OBS
D	truth blocked	OBS explains; no waive
E	interviewing + prepare goal	OBS brief-first
F	illegal delegation	deny visible
G	resume	no duplicate child prep
H	audit reconstruction	parent → handoff → child
I	safety	no pipeline/submit APIs

10. Tests

Suite	Result
tests/unit/multi_agent/ (incl. test_cli_m3.py)	green
tests/unit/agent/test_cli.py	green
M2 corpus (via runtime tests)	green

11. Documentation updated

- This M3 eval
- Implementation notes / changelog / roadmap / functional spec pointers
- ADR-008 milestone note
- Repository navigation (agent help text)

12. Remaining limitations

- Live `orchestrate run` needs a real Opportunity corpus (same as `cic agent`)
- No product dashboard / chat
- No FR-017 eval harness

- Happy-path prep still better via `cic agent run`
 - M4 documentation freeze completed after this milestone ([fr016 m4 evaluation.md](#); [fr016 multi agent orchestration.md](#))
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13. Final repository status

Item	Status
M2 go/no-go	GO AS LEARNING PROOF ONLY (unchanged)
M3 owner CLI	Complete
M4	Not started
FR-017	Not started
Daily prep recommendation	<code>cic agent run</code>

Owner next step

Acknowledge M3. Request M4 only when ready to freeze FR-016 documentation for the learning-proof close-out (still without product-value claims).